Quantum Optics Lecture Notes

Joel Corney

updated

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Main Sources

In preparing these notes, we have relied on the following texts, which you are encouraged to consult for more information:

- P. Meystre and M. Sargent, "Elements of Quantum Optics", 4th ed, Springer, 2007. (M&S)
- M. Orszag, "Quantum Optics", 2nd ed, Springer, 2008.
- M. Fox, "Quantum Optics: an introduction", Oxford University Press, 2006.
- D. F. Walls and G. J. Milburn, "Quantum Optics", 2nd Ed, Springer, 2008.
- G. R. Fowles,"Introduction to Modern Optics", 2nd ed, Dover, 1975.

Quantized Light

Quantum States of Light

1 EM field as a set of quantum harmonic oscillators

Further reading: M&S13.1-2

To go beyond the semiclassical approach, we need to quantize the electromagnetic field itself. As we shall see, the concept of photons appears naturally (and unavoidably). We will also find that, as for the quantum mechanics of particles, so for quantum fields there are canonically conjugate variables whose simultaneous values cannot be predicted with absolute certainty. In other words, the quantized light has an irreducible uncertainty, or quantum noise.

For example, in quantum mechanics, the position and momentum of a particle are incompatible observables, represented by noncommuting operators. In electromagnetism, the electric field is determined through the placement of a test charge, whereas to determine the magnetic field, we also need to know the velocity of the test charge: $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$. So it is no surprise to find out that we cannot determine both the electric and magnetic field amplitudes exactly, and that there are some situations where $\mathbf{E}$ and $\mathbf{B}$ behave like conjugate variables, with an associated uncertainty relation.

Given that light is an oscillatory (wave) solution of Maxwell's equations, it is not surprising that the route to quantising an optical field is to treat is as a set of harmonic oscillators. The general approach will be to determine the oscillating modes in the classical problem, express each mode in the form of a harmonic oscillator, and then quantise it accordingly.

We begin by quantise a single standing wave in a cavity before treating the more general, multimode problem

1.1 Quantizing a single optical mode

Consider a standing wave in a cavity whose electric field, polarised in the x direction, has the form

$$\mathbf{E}(z,t) = \hat{\mathbf{x}} q(t) \sqrt{\frac{2\Omega^2}{\epsilon_0 V}} \sin Kz, \tag{1}$$

where K is the wave number and V is the mode volume. For this field to obey the wave equation

$$\nabla^2\mathbf{E} - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\mathbf{E} = 0, \tag{2}$$

the frequency must be $\Omega = cK$ and the time-dependent amplitude must obey the SHO equation $\ddot{q} + \Omega^2 q = 0$. The corresponding magnetic field is

$$\mathbf{B}(z,t) = \hat{\mathbf{y}}\frac{\dot{q}(t)}{c\Omega}\sqrt{\frac{2\Omega^2}{\epsilon_0 V}}\cos Kz, \tag{3}$$

giving the total energy

$$\begin{aligned}\mathcal{H} &= \frac{\epsilon_0}{2}\int dV\left(E^2 + c^2B^2\right) \\ &= \frac{1}{2}\left(\Omega^2 q^2 + p^2\right),\end{aligned} \tag{4}$$

where the momentum is defined as $p = \dot{q}$. We recognise this as the hamiltonian for a harmonic oscillator, with the energy oscillating back and forth between q (the electric field) and p (the magnetic field). That p is the conjugate variable to q can be confirmed by checking that the Hamilton-Jacobi equations of motion:

$$\begin{aligned}\dot{q} &= \frac{\partial}{\partial p}\mathcal{H} = p \\ \dot{p} &= -\frac{\partial}{\partial q}\mathcal{H} = -\Omega^2 q\end{aligned} \tag{5}$$

satisfy the Maxwell-equation requirement $\ddot{q} + \Omega^2 q = 0$.

Quantization now proceeds by imposing canonical commutation relations on the position and momentum variables:

$$[\hat{q}, \hat{p}] = i\hbar, \tag{6}$$

so that the Heisenberg equations of motion:

$$\begin{aligned}\dot{\hat{q}} &= \frac{1}{i\hbar}\left[\hat{q}, \hat{\mathcal{H}}\right] = \hat{p} \\ \dot{\hat{p}} &= \frac{1}{i\hbar}\left[\hat{p}, \hat{\mathcal{H}}\right] = -\Omega^2\hat{q}\end{aligned} \tag{7}$$

have the same form as the Hamilton-Jacobi equations.

It is very handy to introduce creation $a^\dagger$ and annihilation a operators:

$$a^\dagger = \frac{1}{\sqrt{2\hbar\Omega}}(\Omega\hat{q} - i\hat{p}); \qquad a = \frac{1}{\sqrt{2\hbar\Omega}}(\Omega\hat{q} + i\hat{p}), \tag{8}$$

or inversely

$$\hat{q} = \sqrt{\frac{\hbar}{2\Omega}}\left(a + a^\dagger\right); \qquad \hat{p} = \sqrt{\frac{\hbar\Omega}{2}}\left(\frac{a - a^\dagger}{i}\right) \tag{9}$$

which put the hamiltonian in the form:

$$\begin{aligned} \hat{\mathcal{H}} &= \frac{1}{2}\hbar\Omega\left(a^\dagger a + a a^\dagger\right) \\ &= \hbar\Omega\left(a^\dagger a + \frac{1}{2}\right). \end{aligned} \tag{10}$$

The commutation relations

$$\left[a, a^\dagger\right] = 1 \tag{11}$$

show that the energy eigenstates are number states $|n\rangle$ with energies given by

$$\mathcal{H}_n = \hbar\Omega\left(n + \frac{1}{2}\right), \quad n = 0,\, 1,\, 2,\, \ldots. \tag{12}$$

We can interpret n as the number of photons, or quanta of excitation, in the field. Note that in this interpretation, a photon belongs to the whole standing wave, rather than being a particle localised in space.

In terms of annihilation and creation operators, the electric field operator, in the Heisenberg picture, is:

$$\begin{aligned} \hat{\mathbf{E}}(z,t) &= \sqrt{\frac{\hbar\Omega}{\epsilon_0 V}}\left(a(t) + a^\dagger(t)\right)\sin(Kz)\hat{\mathbf{x}} \\ &= \sqrt{\frac{\hbar\Omega}{\epsilon_0 V}}\left(a e^{-i\Omega t} + a^\dagger e^{i\Omega t}\right)\sin(Kz)\hat{\mathbf{x}} \\ &= \hat{\mathbf{E}}^+(z,t) + \hat{\mathbf{E}}^-(z,t), \end{aligned} \tag{13}$$

i.e. the positive frequency part of the field is an annihilation operator and the negative frequency part is a creation operator. The field amplitude can be defined as:

$$\hat{\mathcal{E}} = \sqrt{\frac{\hbar\Omega}{\epsilon_0 V}}a, \tag{14}$$

and loosely speaking, we can describe $\sqrt{\hbar\Omega/\epsilon_0 V}$ as the electric field per photon. Note, however, that the expectation value of electric field in a number state is zero, essentially because the energy depends on the magnitude of the electric field, not

the phase. The square of the field, on the other hand, has the expectation value

$$\begin{aligned}\langle n|\hat{E}^2|n\rangle &= \frac{\hbar\Omega}{\epsilon_0 V}\langle n|\left(a(t)+a^\dagger(t)\right)^2|n\rangle \sin^2(Kz) \\ &= \frac{\hbar\Omega}{\epsilon_0 V}\langle n|\left(a^\dagger a + aa^\dagger\right)|n\rangle \sin^2(Kz) \\ &= \frac{\hbar\Omega}{\epsilon_0 V}(2n+1)\sin^2(Kz). \end{aligned} \tag{15}$$

Averaging over the mode volume, we obtain:

$$\begin{aligned}\overline{E^2} &= \frac{1}{V}\int dV \langle n|E^2|n\rangle \\ &= \frac{\hbar\Omega}{\epsilon_0 V}\left(n+\frac{1}{2}\right), \end{aligned} \tag{16}$$

i.e. each photon contributes $\hbar\Omega/\epsilon_0 V$ to the mean squared field.

The vacuum state $|0\rangle$ for the mode is defined as the state with no photons, such that $a|0\rangle = 0$. Even though it is the ground state, it has some energy $\hbar\Omega/2$ and hence also a nonzero square of the electric field $\overline{E^2} = \hbar\Omega/2\epsilon_0 V$. One can think of the vacuum as a randomly fluctuating electric field with zero mean but with energy of half a virtual photon per mode. Since this vacuum field cannot be detected by an absorption measurement (there are no real photons to absorb), it is often convenient to subtract the vacuum contribution off the energy, which is equivalent to writing the original Hamiltonian in normal order, i.e. with annihilation operators to the right

$$\begin{aligned}:\hat{\mathcal{H}}: &= \frac{1}{2}\hbar\Omega :\left(a^\dagger a + aa^\dagger\right): \\ &= \frac{1}{2}\hbar\Omega\left(a^\dagger a + a^\dagger a\right) \\ &= \hbar\Omega a^\dagger a \end{aligned} \tag{17}$$

1.2 Quantizing a multimode field

A multimode field in a cavity can be quantized along similar lines, with each mode m acting as a harmonic oscillator with its own annihilation and creation operators a_m, $a_m^\dagger$, with commutation relations

$$\left[a_m, a_{m'}^\dagger\right] = \delta_{m,m'}. \tag{18}$$

The total hamiltonian is then

$$\hat{\mathcal{H}} = \sum_m \hbar\Omega_m\left(a_m^\dagger a_m + \frac{1}{2}\right). \tag{19}$$

Notice that, since there is no limit to the number of modes, the energy of the vacuum is infinite!

The energy eigenstates are now multimode number states,

$$\begin{aligned} |\{n\}\rangle &\equiv |n_1, n_2, \ldots, n_m, \ldots\rangle \\ &\equiv |n_1\rangle \otimes |n_2\rangle \otimes \cdots \otimes |n_m\rangle \otimes \ldots \end{aligned} \tag{20}$$

The modal decomposition need not be performed using standing waves, and for example, in free space, an expansion in terms of running waves is appropriate. In terms of general normalised mode functions $\mathbf{u}_m(\mathbf{r})$, the electric field is

$$\begin{aligned} \hat{\mathbf{E}}(\mathbf{r},t) &= \sum_m \sqrt{\frac{\hbar\Omega_m}{2\epsilon}} \left[\mathbf{u}_m(\mathbf{r}) a_m(0) e^{-i\Omega_m t} + \mathbf{u}_m^*(\mathbf{r}) a_m^\dagger(0) e^{i\Omega_m t}\right] \\ &= \hat{\mathbf{E}}^+(\mathbf{r},t) + \hat{\mathbf{E}}^-(\mathbf{r},t), \end{aligned} \tag{21}$$

where the positive-frequency part of the field is seen to be a multimode annihilation operator.

It is sometimes useful to work with the vector potential operator, which in the radiation gauge is related to the electric field by $\mathbf{E} = -\partial\mathbf{A}/\partial t$:

$$\hat{\mathbf{A}}(\mathbf{r},t) = \sum_m -i\sqrt{\frac{\hbar}{2\epsilon\Omega_m}} \left[\mathbf{u}_m(\mathbf{r}) a_m(0) e^{-i\Omega_m t} - \mathbf{u}_m^*(\mathbf{r}) a_m^\dagger(0) e^{i\Omega_m t}\right]. \tag{22}$$

The magnetic field operator can now be obtained:

$$\begin{aligned} \hat{\mathbf{B}}(\mathbf{r},t) &= \nabla \times \hat{\mathbf{A}} \\ &= \sum_m -i\sqrt{\frac{\hbar}{2\epsilon\Omega_m}} \left[\{\nabla \times \mathbf{u}_m(\mathbf{r})\}\, a_m(0) e^{-i\Omega_m t} - \{\nabla \times \mathbf{u}_m^*(\mathbf{r})\}\, a_m^\dagger(0) e^{i\Omega_m t}\right], \end{aligned} \tag{23}$$

1.3 Plane-wave basis

For a quantizing a field in free space, we work in a basis of running plane-wave modes:

$$\mathbf{u}_m = \frac{1}{\sqrt{V}} e^{i\mathbf{k}_m\cdot\mathbf{r}} \hat{\mathbf{e}}_m, \tag{24}$$

where $V = L^3$ is the quantisation volume, $\hat{\mathbf{e}}_m$ is a unit vector describing the polarisation, and $\mathbf{k}_m$ are the allowed wavevectors:

$$\mathbf{k}_m = \frac{2\pi}{L}(m_x\hat{\mathbf{i}} + m_y\hat{\mathbf{j}} + m_z\hat{\mathbf{k}}), \tag{25}$$

for integers m_x, m_y, m_z. Gauss's law $\nabla\cdot\mathbf{E} = 0$ leads to the transversality condition:

$$\hat{\mathbf{e}}_m \cdot \mathbf{k}_m = 0, \tag{26}$$

which means that for every spatial mode there are two possible polarisation modes, perpendicular to the wavevector.

In this basis, the electric field, vector potential and magnetic field are:

$$\begin{aligned}
\hat{\mathbf{E}}(\mathbf{r},t) &= \sum_m \sqrt{\frac{\hbar\Omega_m}{2\epsilon V}} \left[a_m e^{i(\mathbf{k}_m\cdot\mathbf{r}-\Omega_m t)} + a_m^\dagger e^{-i(\mathbf{k}_m\cdot\mathbf{r}-\Omega_m t)}\right] \hat{\mathbf{e}}_m \\
\hat{\mathbf{A}}(\mathbf{r},t) &= \sum_m -i\sqrt{\frac{\hbar}{2\epsilon\Omega_m V}} \left[a_m e^{i(\mathbf{k}_m\cdot\mathbf{r}-\Omega_m t)} - a_m^\dagger e^{-i(\mathbf{k}_m\cdot\mathbf{r}-\Omega_m t)}\right] \hat{\mathbf{e}}_m \\
\hat{\mathbf{B}}(\mathbf{r},t) &= \sum_m \sqrt{\frac{\hbar\Omega_m}{2\epsilon c^2 V}} \left[a_m e^{i(\mathbf{k}_m\cdot\mathbf{r}-\Omega_m t)} + a_m^\dagger e^{-i(\mathbf{k}_m\cdot\mathbf{r}-\Omega_m t)}\right] \hat{\mathbf{k}}_m \times \hat{\mathbf{e}}_m.
\end{aligned} \tag{27}$$

Using the commutation relations for the annihilation and creation operators and the orthonormality of the mode functions, we find the commutator between the electric field and vector potential to be:

$$\left[\hat{A}_i(\mathbf{r}), \hat{E}_j(\mathbf{r}')\right] = \frac{-i\hbar}{2V\epsilon_0} \sum_{l,\sigma} \left(\hat{e}_{l,\sigma}\right)_i \left(\hat{e}_{l,\sigma}\right)_j \left[e^{i\mathbf{k}_l\cdot(\mathbf{r}-\mathbf{r}')} + e^{-i\mathbf{k}_l\cdot(\mathbf{r}-\mathbf{r}')}\right], \tag{28}$$

where i, j label the cartesian components, l labels the spatial mode and σ labels the polarisation mode. The spatial part of this expression behaves like a Dirac delta function:

$$\lim_{V\to\infty} \frac{1}{V} \sum_l e^{-i\mathbf{k}_l\cdot\mathbf{r}} = \frac{1}{(2\pi)^3} \int e^{-i\mathbf{k}\cdot\mathbf{r}} d^3k = \delta(\mathbf{r}). \tag{29}$$

The polarisation part behaves something like a Kronecker delta function:

$$\sum_\sigma \left(\hat{e}_{l,\sigma}\right)_i \left(\hat{e}_{l,\sigma}\right)_j = \hat{e}_{1i}\hat{e}_{1j} + \hat{e}_{2i}\hat{e}_{2j} = \delta_{ij} - \hat{k}_i\hat{k}_j, \tag{30}$$

which follows from the fact that $\hat{\mathbf{e}}_1$ and $\hat{\mathbf{e}}_2$ are orthogonal to $\hat{\mathbf{k}}$. If we only ever deal with transverse fields, then this reduces to the normal Kronecker delta function. Combining the two results together, we can write the commutator in terms of the 'transverse delta function'[1] $\delta^T_{ij}(\mathbf{r})$:

$$\left[\hat{A}_i(\mathbf{r}), \hat{E}_j(\mathbf{r}')\right] = \frac{-i\hbar}{\epsilon_0} \delta^T_{ij}(\mathbf{r}-\mathbf{r}'), \tag{31}$$

[1] A vector field can be written as the sum $\mathbf{X} = \mathbf{X}_T + \mathbf{X}_L$ of a curl-free part $\nabla\times\mathbf{X}_L = 0$ and a divergence-free part $\nabla\cdot\mathbf{X}_T = 0$. The transverse delta function extracts the transverse part of the field: $\int \delta^T(\mathbf{r}-\mathbf{r}')\cdot\mathbf{X}(\mathbf{r}')d^3r' = \mathbf{X}_T(\mathbf{r})$.

which shows that, in general, $\hat{\mathbf{A}}$ and $\epsilon_0\hat{\mathbf{E}}$ are conjugate fields that cannot be simultaneously specified.

2 Coherent states

Further reading: M&S13.4

Having formally introduced the photon, we now look at its role in two very useful kinds of quantum states: *thermal* (or *chaotic*) states and *coherent* states. These states have a classical interpretation, but quantum formulation give us a 'photon' description that combines the wave and particle pictures.

In the following we will just consider the state of a single-mode field (which could be e.g. a standing wave in a cavity, or a propagating plane wave in free space), but the results are easily generalised to the multimode situation.

2.1 Quadratures and phase space

Coherent states have both a physical significance (as fully coherent light, or an idealisation of laser output) and a mathematical utility (as a basis set for representing a general quantum state). They were first constructed by Schrödinger in 1926, and ever since Glauber utilized them in 1963 in his quantum theory of coherence, they have played a central role in quantum optics.

Classically, a coherent state is characterised by a well defined amplitude $\mathcal{E}$. In quantum mechanics, the uncertainty principle prohibits us from specifying $\mathcal{E}$ exactly, because the real and imaginary parts of $\mathcal{E}$ correspond to noncommuting operators, but we can define a state that has an amplitude with the smallest variance possible. This smallest variance is the uncertainty of the vacuum state.

To quantify these statements, we introduce the *quadrature* operators:

$$\begin{aligned} X_1 &= (a^\dagger + a) \\ X_2 &= i(a^\dagger - a), \end{aligned} \tag{32}$$

which, up to a constant factor, are just the 'position' and 'momentum' operators introduced earlier in the Hamiltonian formulation[2] (see Eq. (9)). Classically, X_1 and X_2 constitute a phase space, in which the system dynamics can be mapped out with deterministic trajectories. Quantumly, the state at a particular time cannot be specified by a single point in phase-space, due to quantum indeterminacy. Instead, a quantum state can be thought of as a distribution of points in phase space, centred at $(\langle X_1\rangle, \langle X_2\rangle)$ and characterised by the variances $\mathrm{var}(X_1)$, $\mathrm{var}(X_2)$.

[2]For a standing wave in a cavity and with the appropriate choice of phase, $\langle X_1\rangle$ is proportional to the instantaneous electric field and $\langle X_2\rangle$ is proportional to the instantaneous magnetic field, which oscillates out of phase. For a running wave in free space, this doesn't hold, as the electric and magnetic fields oscillate in phase. In this case, $\langle X_1\rangle$ gives the $\cos Kz$ contribution to the electric field and $\langle X_2\rangle$ gives the $\sin Kz$ contribution

For example, the vacuum state, for which

$$\begin{aligned} \langle X_1 \rangle &= \langle 0|a + a^\dagger|0\rangle \\ &= 0 = \langle X_2 \rangle \\ \langle X_1^2 \rangle &= \langle 0|aa + aa^\dagger + a^\dagger a + a^\dagger a^\dagger|0\rangle \\ &= 1 = \langle X_2^2 \rangle, \end{aligned} \tag{33}$$

corresponds to a circular distribution centred at the origin with standard deviation in any direction of 1.

2.2 Coherent displacements

To generate the coherent state, we want to add energy to system in a coherent fashion without increasing the noise. Mathematically we can describe this with the displacement operator $\hat{D}(\alpha)$:

$$|\alpha\rangle \equiv \hat{D}(\alpha)|0\rangle \equiv e^{\alpha a^\dagger - \alpha^* a}|0\rangle, \tag{34}$$

where the complex number α is called the coherent amplitude. Some handy properties of the displacement operator include

$$\begin{aligned} \hat{D}^\dagger(\alpha) a \hat{D}(\alpha) &= a + \alpha \\ \hat{D}^\dagger(\alpha) a^\dagger \hat{D}(\alpha) &= a^\dagger + \alpha^* \\ \hat{D}^\dagger(\alpha) &= \hat{D}^{-1}(\alpha) \\ &= D(-\alpha). \end{aligned} \tag{35}$$

from which it follows that:

$$\begin{aligned} \langle \alpha|X_1|\alpha\rangle &= \frac{1}{2}\langle 0|\hat{D}^\dagger(\alpha)(a + a^\dagger)\hat{D}(\alpha)|0\rangle \\ &= \frac{1}{2}\langle 0|(a + \alpha + a^\dagger + \alpha^*)|0\rangle \\ &= \Re(\alpha) \\ \langle \alpha|X_2|\alpha\rangle &= \Im(\alpha) \\ \langle \alpha|X_1^2|\alpha\rangle &= \Re(\alpha)^2 + 1 \\ \langle \alpha|X_2^2|\alpha\rangle &= \Im(\alpha)^2 + 1. \end{aligned} \tag{36}$$

Thus the state is now centred at the point $(\langle X_1\rangle, \langle X_2\rangle) = (\Re(\alpha), \Im(\alpha))$, but with the same noise properties as the vacuum. Or equivalently we can say the coherent state defined in this way is just the vacuum state 'displaced' in the complex plane

by the amplitude α. Furthermore, the energy of the coherent state is:

$$\begin{aligned}\langle \mathcal{H} \rangle &= \frac{1}{2}\hbar\Omega\langle a^\dagger a + aa^\dagger\rangle \\ &= \frac{1}{4}\hbar\Omega\left(\langle X_1^2\rangle + \langle X_2^2\rangle\right) \\ &= \frac{1}{4}\hbar\Omega\left(|\alpha|^2 + \frac{1}{2}\right), \end{aligned} \tag{37}$$

i.e. we have added an average of $\langle n\rangle = |\alpha|^2$ photons to the vacuum without adding any additional noise.

2.3 Coherent state properties

Coherent states:

- are eigenstates of the annihilation operator, with eigenvalue α:

$$\begin{aligned} a|\alpha\rangle &= a\hat{D}(\alpha)|0\rangle \\ &= D(\alpha)D^\dagger(\alpha)a\hat{D}(\alpha)|0\rangle \\ &= D(\alpha)(a+\alpha)|0\rangle \\ &= D(\alpha)\alpha|0\rangle \\ &= \alpha|\alpha\rangle. \end{aligned} \tag{38}$$

- are not eigenstates of the creation operator $a^\dagger|\alpha\rangle \neq \alpha^*|\alpha\rangle$, and are therefore not energy eigenstates $\hat{n}|\alpha\rangle \neq n|\alpha\rangle$.

- have an indefinite number of photons, with variance

$$\begin{aligned} \mathrm{var}(n) &= \langle n^2\rangle - \langle n\rangle^2 \\ &= \langle\alpha|a^\dagger a^\dagger aa + a^\dagger a|\alpha\rangle - \left(\langle\alpha|a^\dagger a|\alpha\rangle\right)^2 \\ &= |\alpha|^4 + |\alpha|^2 - |\alpha|^4 = \overline{n}, \end{aligned} \tag{39}$$

  where $\overline{n} \equiv |\alpha|^2$, the average number of photons.

- have an phase uncertainty that decreases with $\overline{n}$:

$$\Delta\phi = \frac{1}{2\sqrt{\overline{n}}}. \tag{40}$$

- are minimum-uncertainty states with

$$\Delta X_1 \Delta X_2 = \frac{1}{2}\left|\langle [X_1, X_2]\rangle\right| = 1, \tag{41}$$

and

$$\Delta n \Delta\phi = \frac{1}{2}. \tag{42}$$

- can be expressed in terms of creation operators as:

$$\begin{aligned} |\alpha\rangle &= e^{\alpha a^\dagger - \alpha^* a}|0\rangle \\ &= e^{\alpha a^\dagger} e^{-\alpha^* a} e^{-|\alpha|^2/2}|0\rangle \\ &= e^{-|\alpha|^2/2} e^{\alpha a^\dagger}|0\rangle \\ &= e^{-|\alpha|^2/2} \sum_n \frac{\alpha^n (a^\dagger)^n}{n!}|0\rangle, \end{aligned} \tag{43}$$

  where we have used the Campbell-Baker-Hausdorff formula:

$$e^{\hat{A}+\hat{B}} = e^{\hat{A}} e^{\hat{B}} e^{-\frac{1}{2}[\hat{A},\hat{B}]}, \tag{44}$$

  which holds when both $\hat{A}$ and $\hat{B}$ commute with $[\hat{A}, \hat{B}]$.

- are superpositions of number states:

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_n \frac{\alpha^n}{\sqrt{n!}}|n\rangle, \tag{45}$$

  since $(a^\dagger)^n|0\rangle = \sqrt{n!}|n\rangle$.

- are not orthogonal:

$$|\langle\alpha|\beta\rangle|^2 = e^{-|\alpha-\beta|^2} \neq \delta(\alpha - \beta), \tag{46}$$

  but if α and β are sufficiently different, the states will be approximately orthogonal.

- form a complete basis[3], with the resolution of the identity given by[4]:

$$\hat{1} = \frac{1}{\pi} \int d^2\alpha |\alpha\rangle\langle\alpha|, \tag{48}$$

  which means any other state can be represented as a superposition of coherent states.

[3] actually, since they are not orthogonal, they are *overcomplete*, which leads to nonuniqueness when it comes to expressing other states as distributions over coherent states

[4] we can prove this by expanding in terms of number states and using the complex gaussian integral:

$$\int d^2\alpha e^{-|\alpha|^2} = \pi \tag{47}$$

- are very useful for evaluating expectation values of normally ordered operator products, since

$$\langle\alpha| : f(a^\dagger, a) : |\alpha\rangle = f(\alpha^*, \alpha). \tag{49}$$

3 Photon statistics and photodetection

Further reading: M&S13.3

3.1 Thermal states

We now turn to a quantum description of the other main paradigm of classical light, namely the light from a thermal source, such as blackbody radiation or the output from an discharge lamp. Again we will focus on a single mode, which describes thermal light that has been spectrally filtered to produced (quasi)monochromatic light.

In quantum mechanics, a thermal ensemble is described by a density operator:

$$\rho = \frac{\exp(-\mathcal{H}/kT)}{\mathrm{Tr}\left[\exp(-\mathcal{H}/kT)\right]}, \tag{50}$$

an expression which can be derived by maximizing the entropy $\mathcal{S} = -k\mathrm{Tr}\left[\rho \ln \rho\right]$, subject to the constraint of fixed energy $\langle \mathcal{H} \rangle = \mathrm{Tr}[\mathcal{H}\rho]$ and normalisation $\mathrm{Tr}[\rho] = 1$.

For the specific case of a single optical mode, with $\mathcal{H} = \hbar\Omega a^\dagger a$, the thermal state is

$$\rho = \frac{\exp(-\hbar\Omega a^\dagger a/kT)}{\mathrm{Tr}\left[\exp(-\hbar\Omega a^\dagger a/kT)\right]}. \tag{51}$$

Using the completeness of thermal states $\hat{1} = \sum_n |n\rangle\langle n|$, we can rewrite this as

$$\begin{aligned} \rho &= \frac{\sum_n e^{-\hbar\Omega a^\dagger a/kT}|n\rangle\langle n|}{\sum_n \langle n|e^{-\hbar\Omega a^\dagger a/kT}|n\rangle} \\ &= \frac{\sum_n e^{-\hbar\Omega n/kT}|n\rangle\langle n|}{\sum_n e^{-\hbar\Omega n/kT}} \\ &= \left[1 - e^{-\hbar\Omega/kT}\right] \sum_n e^{-\hbar\Omega n/kT}|n\rangle\langle n|, \end{aligned} \tag{52}$$

i.e. a mixture of number states with no coherences (off-diagonal elements) between them. The diagonal elements of a density operator give the probabilities of being in each state, which in this case just corresponds to the Boltzmann result:

$$\begin{aligned} p_n &= \left[1 - e^{-\hbar\Omega/kT}\right] e^{-\hbar\Omega n/kT} \\ &= \frac{1}{1+\overline{n}} \left(\frac{\overline{n}}{1+\overline{n}}\right)^n, \end{aligned} \tag{53}$$

where $\overline{n}$ is the average number of photons:

$$\overline{n} = \frac{1}{e^{\hbar\Omega/kT} - 1}. \tag{54}$$

The variance of the thermal photon number distribution is

$$\mathrm{var}(n) = \overline{n} + \overline{n}^2. \tag{55}$$

3.2 Poissonian statistics

For a coherent state, the distribution of photon number is

$$\begin{aligned} p_n &= |\langle n|\alpha\rangle|^2 \\ &= e^{-|\alpha|^2}\frac{|\alpha|^{2n}}{n!} \\ &= e^{-\overline{n}}\frac{\overline{n}^n}{n!}, \end{aligned} \tag{56}$$

which is the *Poisson* distribution, characterised by a variance equal to the mean $\mathrm{var}(n) = \overline{n}$. A thermal distribution with the same mean will always have a larger variance than the coherent case: $\mathrm{var}(n) = \overline{n} + \overline{n}^2$. Furthermore, the exponential form of the thermal distribution is always highly skewed, with $n = 0$ the most likely value, whereas the Poisson distribution is peaked around its mean, approaching a symmetrical gaussian for large $\overline{n}$.

The photon statistics can be directly measured experimentally if we have a photodetector that can respond to individual photons, creating a pulse of current for each absorption event. In practice this can be achieved for low photon flux rates (less than 10^6 photons per second) by use of a photomultiplier tube or an avalanche photodiode (APD). An analysis of the mean and the variance of the number of detection events in a given period will then tell us about the type of statistics of the light[5]. To classify the results, we use following terminology:

superpoissonian $\mathrm{var}(n) > \overline{n}$, e.g. a thermal state. Observation of poissonian statistics can be explained (as we shall see in the next section) with a semiclassical model that includes intensity fluctuations.

poissonian $\mathrm{var}(n) = \overline{n}$, e.g. a coherent state. In a semiclassical model this is explained by a light of constant intensity.

subpoissonian $\mathrm{var}(n) < \overline{n}$, i.e. smaller number fluctuations than a coherent state, e.g. a number state. Observation of subpoissonian statistics has no semiclassical explanation and thus provides a clear demonstration that it is light itself that is quantized rather than just the detection process.

[5] assuming that every photon leads to an electrical pulse, i.e. perfect quantum efficiency

Here we have gone from a picture of photons as quanta of excitations in a delocalised mode to a picture of photons as localised entities that hit the photoactive region of the detector at particular points in time. We can go from one picture to the other by thinking of the photons that are absorbed in the time interval t as belonging to a mode of quantisation length $L = ct$.

3.3 Semiclassical Theory of photodetection

We can develop a semiclassical model to explain the statistics of photodetection based on a few simple assumptions:

1. The rate at which elections in the photoactive region are excited depends only on the intensity of the incident light $I(t)$. In a sufficiently small amount of time Δt, probability of one electron being excited is therefore
$$P_1(t, t + \Delta t) = \xi I(t)\Delta t, \tag{57}$$
where ξ is some constant.

2. We can choose Δt to be small enough such that the probability of two detection events in the time Δt is negligible. The probability of no detection events is thus
$$P_0(t, t + \Delta t) = 1 - \xi I(t)\Delta t. \tag{58}$$

3. A detection event does not affect the probability of subsequent detections (this gives a practical limit to the photon flux, since a detector often has a 'deadtime' after each detection event during which the detector cannot respond to further absorptions)

To determine the probability for n detection events over a longer time period, we develop a recursion relation as follows. The probability $P_n(0, t + \Delta t)$ can be written as the probability of n events in the interval $0 \to t$ and 0 in the interval $t \to t + \Delta t$ plus the probability of $n - 1$ events in the interval $0 \to t$ and 1 in the interval $t \to t + \Delta t$. Using the independence assumption above, we can write this as:

$$\begin{aligned} P_n(0, t + \Delta t) &= P_n(0, t)P_0(t, t + \Delta t) + P_{n-1}(0, t)P_1(t, t + \Delta t) \\ &= P_n(0, t)\left[1 - \xi I(t)\Delta t\right] + P_{n-1}(0, t)\xi I(t)\Delta t. \end{aligned} \tag{59}$$

In the limit that $\Delta t \to 0$, the rate at which the probability grows with detection time is:

$$\begin{aligned} \frac{dP_n}{dt} &= \frac{P_n(t + \Delta t) - P_n(t)}{\Delta t} \\ &= \xi I(t)\left[P_{n-1}(t) - P_n(t)\right]. \end{aligned} \tag{60}$$

The solution to this set of coupled ODE's is (check by substitution):

$$P_n(t) = \frac{\left[\int_0^t \xi I(t')dt'\right]^n}{n!} \exp\left[-\int_0^t \xi I(t')dt'\right] \tag{61}$$

3.3.1 Constant Intensity

For a constant intensity, the integral is just:

$$\int_0^t \xi I(t')dt = \xi It, \tag{62}$$

which is the same for every measurement interval of length t. The probability of detecting n photons in the interval is then given by a Poisson distribution:

$$P_n(t) = \frac{[\xi It]^n}{n!} \exp[-\xi It], \tag{63}$$

with mean $\overline{n} = \xi It$.

A photocount with poissonian statistics can thus be explained by a semiclassical model with light of constant intensity. In this model, the randomness does not lie with the light (random arrival times of photons) but with the detection process itself.

3.3.2 Fluctuating Intensity

For a fluctuating intensity, the photocount in a particular interval depends on the average current in that interval:

$$\int_0^t \xi I(t')dt' = \xi \overline{I}_t t, \tag{64}$$

where $\overline{I}_t$ denotes a time-average of the current over the interval of length t. This is to be distinguished from an ensemble average, or an average of a set of independent measurements, which we will denote in this section by $\langle \dots \rangle$.

The time-averaged current $\overline{I}_t$ will vary from measurement to measurement. We can treat it as a random variable governed by some distribution $p(\overline{I}_t)$. The photocount formula now needs to be averaged over all possible realisations:

$$P_n(t) = \int \frac{\left[\xi \overline{I}_t t\right]^n}{n!} \exp\left[-\xi \overline{I}_t t\right] p(\overline{I}_t) d\overline{I}_t \tag{65}$$

The mean count is now:

$$\begin{aligned}\langle n\rangle &= \sum_n nP_n(t)\\ &= \int \sum_n n\frac{\left[\xi\overline{I}_t t\right]^n}{n!}\exp\left[-\xi\overline{I}_t t\right]p(\overline{I}_t)d\overline{I}_t\\ &= \int \xi\overline{I}_t t p(\overline{I}_t)d\overline{I}_t\\ &= \xi\langle\overline{I}_t\rangle t \end{aligned} \tag{66}$$

Similarly, the mean square is

$$\begin{aligned}\langle n^2\rangle &= \sum_n n^2P_n(t)\\ &= \int \left[\xi^2\overline{I}_t^2t^2+\xi\overline{I}_t t\right]p(\overline{I}_t)d\overline{I}_t\\ &= \xi^2\langle\overline{I}_t^2\rangle t^2+\xi\langle\overline{I}_t\rangle t, \end{aligned} \tag{67}$$

which leads to the variance:

$$\mathrm{var}(n) = \xi^2\mathrm{var}(\overline{I}_t)t^2+\langle n\rangle. \tag{68}$$

Thus the photocount statistics have a Poissonian contribution from the random nature of the detection process, plus an additional contribution from the intensity fluctuations. A classical field with intensity fluctuations is therefore sufficient to explain superpoissonian statistics.

3.4 Photocurrent fluctuations *(optional)*

It is not always practical, or possible, to determine the photon statistics with single-photon detection. Besides the requirement of low flux rates, another issue is the low quantum efficiency $\eta \sim 0.5$, i.e. only about half of the photons (or absorption events) lead to detectable pulses of current. If we are willing to forego the requirement of being able to resolve every photon, we can use photodiodes to analyse the statistics of the light. In these devices, each photon creates an electron-hole pair that contributes to the current across a reverse-bias diode. The advantages are:

- photodiodes can cope with high flux e.g. $\Phi = 3\times10^{15}$ photons/sec
- They have high quantum efficiency $\eta \simeq 0.95$, meaning that nearly every photon absorbed liberates an electron that contributes to the current. The statistics of the current can then be used to determine the statistics of the light.

The observed photocurrent will be

$$i = \eta e \Phi = \eta e \frac{P}{\hbar \nu}, \tag{69}$$

where P is the incident optical power. If the Δt is the time period over which we can resolve current fluctuations, then the mean and the variance of the current are:

$$\langle i \rangle = e \frac{\langle \eta n \rangle}{\Delta t} \tag{70}$$

$$\mathrm{var}(i) = e^2 \frac{\mathrm{var}(\eta n)}{\Delta t^2}, \tag{71}$$

where n is the number of electrons in time interval Δt.

If the absorption statistics are poissonian, then

$$\begin{aligned} \mathrm{var}(i) &= e^2 \frac{\langle \eta n \rangle}{\Delta t^2} \\ &= \frac{e \langle i \rangle}{\Delta t}, \end{aligned} \tag{72}$$

which means that as $\Delta t \to 0$, the current fluctuations get very large. The solution to this is just to look at fluctuations over a narrow band of frequencies. One can then show that the resultant variance is

$$\mathrm{var}(i) = 2e \Delta f \langle i \rangle, \tag{73}$$

where Δf is the frequency range.

This variance defines the *shot noise* limit, since it arises from the fact that the current is made up of discrete electrons, excited by (in the classical explanation) light of constant intensity. Light with intensity fluctuations will create a current with fluctuations above this level (superpoissonian), whereas current fluctuations below shot noise are possible only with nonclassical subpoissonian light.

4 Quantum coherence and correlations

Further reading: M&S13.5

4.1 The key idea: measurement by absorption

We now wish to place the discussion of photodetection and our earlier discussion of correlation functions on a quantum-mechanical footing. In either case we are concerned with measuring the intensity of the light, and correlations in that intensity. If the atoms in our detectors are in the ground state (a very safe assumption for optical transitions at room temperature), then the only way for the detector to respond to the intensity of the light is for the atom to absorb a photon[6]. The signal we get will be proportional to the rate of transition, which, according to perturbation theory, depends on the matrix element between initial and final states of the system:

$$w_{i\to f} \propto \left|\langle f|\hat{E}^+(x)|i\rangle\right|^2, \tag{74}$$

where $E^+(x) = E^+(\mathbf{r},t)$, the positive-frequency component of the field, is the part of the field operator that contains only annihilation operators:

$$\hat{E}^+(\mathbf{r},t) = \sum_m \sqrt{\frac{\hbar\Omega_m}{2\epsilon}} u_m(\mathbf{r}) a_m(0) e^{-i\Omega_m t}, \tag{75}$$

where we have ignored polarisation.

Since we don't care about the final state of the system after the photon has been absorbed, we sum over all possible final states:

$$\begin{aligned} w &\propto \sum_f \left|\langle f|\hat{E}^+(x)|i\rangle\right|^2 \\ &= \sum_f \langle i| \left[\hat{E}^+(x)\right]^\dagger |f\rangle\langle f|\hat{E}^+(x)|i\rangle \\ &= \sum_f \langle i|\hat{E}^-(x)|f\rangle\langle f|\hat{E}^+(x)|i\rangle \\ &= \langle i|\hat{E}^-(x)\hat{E}^+(x)|i\rangle, \end{aligned} \tag{76}$$

where we have used the completeness relation $1 = \sum_f |f\rangle\langle f|$. Thus the rate of absorbing a photon is a measure of the intensity of light as defined by $\langle \hat{E}^-(x)\hat{E}^+(x)\rangle$.

[6] i.e. only the part of the field with annihilation operators is resonant

In the case that the light is in a mixed state, we calculate this expectation value by use of the density operator ρ for the light:

$$w \propto \mathrm{Tr}\left[\rho \hat{E}^-(x)\hat{E}^+(x)\right]. \tag{77}$$

To appreciate the difference between $\langle \hat{E}^-(x)\hat{E}^+(x)\rangle$ and $\frac{1}{2}\langle \hat{E}(x)\hat{E}(x)\rangle$, as a measure of intensity, consider the 'intensity' of the vacuum:

$$\begin{aligned}
\langle 0|\hat{E}^-(x)\hat{E}^+(x)|0\rangle &= \sum_m \frac{\hbar\Omega_m}{2\epsilon}|u_m(\mathbf{r})|^2\langle 0|a_m^\dagger a_m|0\rangle \\
&= 0 \qquad (78)\\
\langle 0|\hat{E}(x)\hat{E}(x)|0\rangle &= \frac{1}{2}\sum_m \frac{\hbar\Omega_m}{2\epsilon}|u_m(\mathbf{r})|^2\langle 0|a_m^\dagger a_m + a_m a_m^\dagger|0\rangle \\
&= \frac{1}{2}\sum_m \frac{\hbar\Omega_m}{2\epsilon}|u_m(\mathbf{r})|^2 \neq 0, \qquad (79)
\end{aligned}$$

i.e. unless a normally ordered correlation function is used, then the vacuum would have a nonzero (infinite) intensity.

If we wish to use two detectors (or one detector at two different times) to measure photon coincidences, then the relevant matrix element is $\langle f|\hat{E}^+(x)\hat{E}^+(x')|i\rangle$. The same reasoning as above shows the transition rate is:

$$w \propto \langle \hat{E}^-(x')\hat{E}^-(x)\hat{E}^+(x)\hat{E}^+(x')\rangle, \tag{80}$$

which is a measure of intensity correlation.

Similarly, the n-fold coincidence rate is

$$w \propto \langle \hat{E}^-(x_1)\ldots\hat{E}^-(x_n)\hat{E}^+(x_n)\ldots\hat{E}^+(x_1)\rangle. \tag{81}$$

4.2 Quantum coherence

The two key conceptual advances in Glauber's theory of coherence are that

- true coherence is not encapsulated merely by what is called 'first-order' coherence. To fully characterise coherence, we need to introduce correlation functions to all orders (as we have done already for classical fields)
- since coherence is manifest in intensity measurements, the correlation functions of a quantum field must written with the operators in normal order.

The nth order correlation function is thus defined as

$$G^{(n)}(x_1 \dots x_n, y_n \dots y_1) \equiv \langle \hat{E}^-(x_1) \dots \hat{E}^-(x_n) \hat{E}^+(y_n) \dots \hat{E}^+(y_1) \rangle, \tag{82}$$

with nth order coherence defined by the condition that all correlation functions up to nth order factorise into products of amplitudes:

$$G^{(j)}(x_1 \dots x_j, y_j \dots y_1) = \mathcal{E}^*(x_1) \dots \mathcal{E}^*(x_j) \mathcal{E}^*(y_j) \dots \mathcal{E}^*(y_1) \;\; \forall j \leq n \tag{83}$$

.

It is often convenient to work with normalised correlation functions:

$$g^{(n)}(x_1 \dots x_n, y_n \dots y_1) \equiv \frac{G^{(n)}(x_1 \dots x_n, y_n \dots y_1)}{\prod_k^n \sqrt{G^{(1)}(x_k, x_k)} \sqrt{G^{(1)}(y_k, y_k)}}, \tag{84}$$

i.e. we divide the unnormalised correlation function by the square root of the product of the intensities at each point. The condition for n-order coherence then reduces to $|g^{(j)}| = 1$ for all $j \leq n$.

For higher-order coherence, we are often just interested in intensity correlations, for which $x_1 = y_1$, etc. The simplified correlation function is then

$$g^{(n)}(x_1 \dots x_n) \equiv g^{(n)}(x_1 \dots x_n, x_n \dots x_1), \tag{85}$$

which is always real.

4.2.1 First order coherence: monochromaticity

A monochromatic (single-mode) field is always coherent to first order, since[7]

$$\begin{aligned} g^{(1)}(t_1, t_2) &= \frac{\langle a^\dagger e^{i\Omega t_1} a e^{-i\Omega t_2} \rangle}{\sqrt{\langle a^\dagger e^{i\Omega t_1} a e^{-i\Omega t_1} \rangle \langle a^\dagger e^{i\Omega t_2} a e^{-i\Omega t_2} \rangle}} \\ &= e^{-i\Omega(t_2 - t_1)}, \end{aligned} \tag{86}$$

whose magnitude is one. This is true regardless of the photon statistics, i.e. first order coherence does not distinguish between coherent, thermal, number state, etc.

For stationary fields, whose correlation functions depend only on the relative coordinate e.g. $g^{(1)}(t, t+\tau) = g^{(1)}(\tau)$, coherence requires the field to be monochromatic. Fields with a small range of frequencies may be partially coherent, with the coherence function $g^{(1)}(\tau)$ approaching zero for $\tau > \tau_c$, the coherent time. The Fourier transform of $g^{(1)}(\tau)$ is the line-shape function $g(\Omega)$, which gives the spectral profile of the light.

[7] for simplicity we will just look at coherence in time, but similar consideration apply to coherence in space

4.2.2 Second-order coherence: Bunching and Antibunching

To distinguish between thermal and coherent light, we need to consider the second-order correlation function:

$$\begin{aligned} g^{(2)}(\tau) &\equiv g^{(2)}(t, t+\tau; t+\tau, t) \\ &= \frac{\langle : \hat{I}(t)\hat{I}(t+\tau) : \rangle}{\langle \hat{I}(t) \rangle \langle \hat{I}(t+\tau) \rangle}, \end{aligned} \tag{87}$$

where the intensity operator is here defined as $\hat{I} = \hat{E}^- \hat{E}^+$. This gives the intensity correlations as measured by Twiss and Hanbury Brown in their stellar intensity interferometry experiments. For sufficiently large τ, then $g^{(2)} \to 1$ for any kind of light. What is interesting is what happens as $\tau \to 0$. In the classical explanation, $g^{(2)}(0) = 1$ only for light of constant intensity; otherwise the intensity fluctuations will lead to $g^{(2)}(0) > 1$.

To obtain the quantum result, consider $g^{(2)}(0)$ for a single-mode state:

$$\begin{aligned} g^{(2)}(0) &= \frac{\langle a^\dagger a^\dagger a a \rangle}{\langle a^\dagger a \rangle \langle a^\dagger a \rangle} \\ &= \frac{\langle a^\dagger a a^\dagger a \rangle - \langle a^\dagger a \rangle}{\langle a^\dagger a \rangle \langle a^\dagger a \rangle} \\ &= 1 + \frac{\mathrm{var}(n) - \overline{n}}{\overline{n}^2} \end{aligned} \tag{88}$$

For a thermal state, we have already seen that $\mathrm{var}(n) = \overline{n}^2 + \overline{n}$. Thus $g^{(2)}(0) = 2$. In the photon interpretation, this means that detecting one photon makes it more likely that another photon will also be detected, i.e. the photons appear to arrive in bunches, a phenomenon known as *bunching*. More generally, one can show that, for multimode thermal light,

$$g^{(2)}(\tau) = 1 + g^{(1)}(\tau). \tag{89}$$

For a coherent state, $\mathrm{var}(n) = \overline{n}$. Thus $g^{(2)}(0) = 1$. This means that the arrival time of each photon is completely random, uninfluenced by whether other photons have also been detected.

For a number state, $\mathrm{var}(n) = 0$. Thus

$$g^{(2)}(0) = 1 - \frac{1}{n} < 1 \tag{90}$$

which means that probability of detecting simultaneous photons is lowered, which will make the arrival times more regular - or spaced out- than that of the coherent case. This phenomenon, known as *antibunching*, has no classical explanation, and provides direct confirmation of the quantum nature of light.

4.2.3 Coherence state: coherence to all orders

Since all the correlation functions are normally ordered, it follows that for single-mode coherent state they will always factorise into products of amplitudes:

$$\begin{aligned} \left|g^{(n)}\right| &= \frac{\langle a^{\dagger n} a^n \rangle}{\langle a^\dagger a \rangle^n} \\ &= \frac{\alpha^{*n}\alpha^n}{|\alpha|^{2n}} \\ &= 1. \end{aligned} \tag{91}$$

i.e. a coherent state is coherent to all orders and thus very deserving of its name.

4.3 Photon counting: Subpoissonian statistics

The quantum theory of photodetection proceeds along the same lines as the semi-classical derivation in Sec. 3.3, except that now the coincidence rate for N detection events involves an Nth-order correlation function.

$$\langle \left[\hat{E}^-(t)\right]^N \left[\hat{E}^+(t)\right]^N \rangle = \langle : \left[\hat{I}(t)\right]^N : \rangle. \tag{92}$$

The quantum expression for the probability for N detection events in the interval $0 \to t$ turns out to be:

$$P_N(t) = \left\langle : \frac{\left[\int_0^t \xi \hat{I}(t')dt'\right]^N}{N!} \exp\left[-\int_0^t \xi \hat{I}(t')dt'\right] : \right\rangle, \tag{93}$$

and for a single-mode field:

$$P_N(t) = \left\langle : \frac{\left[\eta(t) a^\dagger a\right]^N}{N!} \exp\left[-\eta(t) a^\dagger a\right] : \right\rangle, \tag{94}$$

where $\eta(t)$ is the probability of detecting one photon in a time t from a one-photon field.

For a coherent state, the expectation value is easy to evaluate:

$$P_N(t) = \frac{\left[\eta(t)|\alpha|^2\right]^N}{N!} \exp\left[-\eta(t)|\alpha|^2\right], \tag{95}$$

which is just a Poisson distribution with mean

$$\overline{N} = \eta|\alpha|^2 = \eta\overline{n}, \tag{96}$$

where $\overline{n}$ is the average number photons in the field. The quantity η is thus the quantum efficiency that tells us the fraction of photons that are detected.

For a more general state, one can expand over number states to find the photocount probability P_N in terms of the photon number distribution p_n of the field. The end result is:

$$P_N = \sum_{n=N}^{\infty} p_n B(n; N, \eta), \tag{97}$$

where

$$B(n; N, \eta) = \frac{n!}{N!(n-N)!} \eta^N (1-\eta)^{n-N} \tag{98}$$

is the Binomial distribution that describes the probability that N of the n photons in the field are detected. Using the properties of the Binomial distribution, one can show that the variance in the photon count is

$$\begin{aligned} \mathrm{var}(N) &= \overline{n}\eta(1-\eta) + \eta^2 \mathrm{var}(n) \\ &= \overline{N}(1-\eta) + \eta^2 \mathrm{var}(n), \end{aligned} \tag{99}$$

which contains a Poissonian contribution from the random detection process in addition to a contribution from the underlying photon statistics. If the quantum efficiency is very high $\eta \simeq 1$, then the statistics of the photocount will faithfully reflect the photon statistics of the field, which can be subpoissonian. But if the quantum efficiency is low, then the photocount will approach the Poisson result, regardless of the photon statistics of the field.

5 Quadratures and squeezed states

Further reading: M&S17.1

5.1 Squeezing

Up until now, the only truly nonclassical states we have described have been the number states, and although there has been much progress in recent years with single-photon sources, number states are very challenging to produce reliably. A common alternative to achieving nonclassical behaviour in light has been through *squeezing*, in which the uncertainty in one variable is 'squeezed' below the coherent-state level. For quadratures (defined in Eq. (32)), this means

$$\langle \Delta X_1^2 \rangle < 1, \text{ or } \langle \Delta X_2^2 \rangle < 1, \tag{100}$$

but not both simultaneously, which would violate Heisenberg's uncertainty principle. Besides revealing nonclassical behaviour, such as antibunching and subpoissonian statistics, squeezed states are of interest as a route to generating entangled states and for practical application in reducing noise in interferometric and other kinds of precision measurements.

The ideal squeezed state is, like the coherent state, a minimum-uncertainty state. But its 'noise distribution' in phase space is not symmetric - it is an ellipse rather than a circle. A squeezed state is characterised by both a complex amplitude $\alpha = \sqrt{\overline{n}}e^{i\theta}$ and a complex squeezing parameter $\zeta = re^{2i\phi}$. The parameter r gives the degree of squeezing (i.e. the eccentricity of the ellipse) while ϕ specifies which quadrature is being squeezed (i.e. the angle of the minor axis of the ellipse). If $\phi = 0$, then

$$\begin{aligned} \langle \Delta X_1^2 \rangle &= e^{-2r} \\ \langle \Delta X_2^2 \rangle &= e^{2r}, \end{aligned} \tag{101}$$

i.e. the X_1 quadrature is squeezed and the X_2 quadrature is antisqueezed. Squeezing can be generated by parametrically amplifying one quadrature and deamplifying the other.

Some special cases:

amplitude squeezing: where the squeezed axis lines up with the coherent displacement, i.e. $\theta = \phi$. Reduced amplitude fluctuations lead to subpoissonian number statistics, which can be directly detected through intensity measurements.

phase squeezing: The squeezed axis perpendicular to the coherent displacement $\theta = \phi \pm 90°$ will lead to reduced phase fluctuations, can can reduce the noise in interferometric measurements.

vacuum squeezing: squeezing without a coherent displacement. A squeezed vacuum corresponds to a randomly fluctuating electric field, but with the fluctuations modulated in a phase-sensitive way.

5.2 The Squeezing Operator

To describe the squeezed state mathematically, we adopt the same approach as for a coherent state: create a 'vacuum' with the desired noise properties and then coherently displace it to the right location in phase space.

The squeezing operator is defined as:

$$\hat{S}(\zeta) \equiv e^{(\zeta^* a^2 - \zeta a^{\dagger 2})/2} \tag{102}$$

and some of its useful properties include:

$$\begin{aligned}
\hat{S}^\dagger(\zeta) &= \hat{S}^{-1}(\zeta) = \hat{S}(-\zeta) \\
\hat{S}(\zeta) a \hat{S}^\dagger(\zeta) &= \mu a + \nu a^\dagger \\
\hat{S}(\zeta) a^\dagger \hat{S}^\dagger(\zeta) &= \mu a^\dagger + \nu^* a \\
\hat{S}^\dagger(\zeta) a \hat{S}(\zeta) &= \mu a - \nu a^\dagger \\
\hat{S}^\dagger(\zeta) a^\dagger \hat{S}(\zeta) &= \mu a^\dagger - \nu^* a
\end{aligned} \tag{103}$$

where $\mu = \cosh r$, $\nu = e^{2i\phi} \sinh r$, such that $\mu^2 - |\nu|^2 = 1$.

An ideal squeezed state is defined as:

$$|\alpha, \zeta\rangle \equiv \hat{D}(\alpha)\hat{S}(\zeta)|0\rangle, \tag{104}$$

where $\hat{D}(\alpha)$ is the displacement operator as defined for coherent states, Eq. (34). According to this definition, we first squeeze the vacuum and then displace it. We now check to see that it has the properties we want:

$$\begin{aligned}
\langle a \rangle &= \langle 0|\hat{S}^\dagger \hat{D}^\dagger a \hat{D} \hat{S}|0\rangle \\
&= \langle 0|\hat{S}^\dagger (a + \alpha) \hat{S}|0\rangle \\
&= \langle 0|(\mu a - \nu a^\dagger + \alpha)|0\rangle \\
&= \alpha,
\end{aligned} \tag{105}$$

which means that the expectation value of the quadratures is the same as for a coherent state:

$$\begin{aligned}
\langle \hat{X}_1 \rangle &= (\alpha + \alpha^*) \\
\langle \hat{X}_2 \rangle &= -i(\alpha - \alpha^*).
\end{aligned} \tag{106}$$

To calculate the variance, we need

$$\begin{aligned}
\langle X^2\rangle &= \langle 0|\hat{S}^\dagger D^\dagger (a+a^\dagger)^2 \hat{D}\hat{S}|0\rangle \\
&= \langle 0|\hat{S}^\dagger (a+a^\dagger+\alpha+\alpha^*)^2 \hat{S}|0\rangle \\
&= \langle 0|(\mu a - \nu a^\dagger + \mu a^\dagger - \nu^* a + \alpha + \alpha^*)^2|0\rangle \\
&= \langle 0|\left[(-\mu\nu + \mu^2 - \mu\nu^* + |\nu|^2)aa^\dagger + (\alpha+\alpha^*)^2\right]|0\rangle \\
&= |\mu-\nu|^2 + (\alpha+\alpha^*)^2, \qquad (107)
\end{aligned}$$

where in going from the 3rd to the 4th line we kept only antinormally ordered products. Using some trigonometric and hyperbolic identities, we find

$$\begin{aligned}
\Delta X_1 &= |\mu-\nu| \\
&= [\mu - |\nu|\cos(2\phi)] \\
&= \sqrt{(e^r \sin\phi)^2 + (e^{-r}\cos\phi)^2}. \qquad (108)
\end{aligned}$$

Similarly,

$$\begin{aligned}
\Delta X_2 &= |\mu+\nu| \\
&= [\mu + |\nu|\cos(2\phi)] \\
&= \sqrt{(e^r \cos\phi)^2 + (e^{-r}\sin\phi)^2}. \qquad (109)
\end{aligned}$$

For a squeezing angle of $\phi = 0$, $\hat{X}_1$ will thus be squeezed and $\hat{X}_2$ antisqueezed, as required.

It is interesting to note the average number of photons in a squeezed state:

$$\begin{aligned}
\langle a^\dagger a\rangle &= \langle 0|\hat{S}^\dagger D^\dagger a^\dagger a \hat{D}\hat{S}|0\rangle \\
&= \langle 0|\hat{S}^\dagger (a^\dagger+\alpha^*)(a+\alpha)\hat{S}|0\rangle \\
&= \langle 0|(\mu a^\dagger - \nu^* a + \alpha^*)(\mu a - \nu a^\dagger + \alpha)|0\rangle \\
&= \langle 0|(-\nu^* a + \alpha^*)(\nu a^\dagger + \alpha)|0\rangle \\
&= |\nu|^2 + |\alpha|^2, \qquad (110)
\end{aligned}$$

which means there are more photons in a squeezed state than in a coherent state with the same coherent amplitude. In particular, the photon number of a 'squeezed vacuum' is nonzero, which arises from the fact that it requires input of energy (a 'pump') to squeeze a state.

5.2.1 A different view of the same squeezed state

We can use the squeezing operator to define new annihilation and creation operators:

$$\begin{aligned}
b &\equiv \hat{S}a\hat{S}^\dagger = \mu a + \nu a^\dagger \\
b^\dagger &\equiv \hat{S}a^\dagger\hat{S}^\dagger = \mu a^\dagger + \nu a, \qquad (111)
\end{aligned}$$

which obey the standard canonical commutation relations:[8]

$$\left[b, b^\dagger\right] = 1. \tag{112}$$

One can show that the squeezed state is an eigenstate of the b operator:

$$b|\alpha, \zeta\rangle = \beta|\alpha, \zeta\rangle, \tag{113}$$

with eigenvalue $\beta \equiv \mu\alpha + \nu\alpha^*$. The squeezed state is thus a coherent state for the transformed operator b, which creates 'squeezed photons' from the squeezed vacuum:

$$|\alpha, \zeta\rangle = \hat{D}_b(\beta)|0\rangle_s, \tag{114}$$

where $|0\rangle_s \equiv \hat{S}|0\rangle$ is the squeezed vacuum. Defined in this way, squeezed states are known as 'two-photon coherent states'.

5.2.2 A definition of a different squeezed state

An alternative way to achieve a squeezed state is to take a coherent state and then squeeze it[9], i.e. swap the order of the squeezing and displacement operations:

$$|\zeta, \alpha\rangle \equiv \hat{S}(\zeta)\hat{D}(\alpha)|0\rangle, \tag{115}$$

which is not equivalent to our original definition. Although the noise properties are the same, the mean values of the quadratures are different, because the squeezing operator parametrically amplifies the the mean of the quadratures as well as the noise. Thus, for example, the coherent amplitude with this new definition is:

$$\begin{aligned} \langle a \rangle &= \langle\alpha|\hat{S}^\dagger a \hat{S}|\alpha\rangle \\ &= \langle\alpha|(\mu a - \nu a^\dagger)|\alpha\rangle \\ &= \mu\alpha - \nu\alpha^* \end{aligned} \tag{116}$$

We can get back to the original definition of the squeezed state if we first displace by $\beta = \mu\alpha + \nu\alpha^*$ rather than α, i.e.

$$|\alpha, \zeta\rangle \equiv \hat{D}(\alpha)\hat{S}(\zeta)|0\rangle = \hat{S}(\zeta)\hat{D}(\beta)|0\rangle \equiv |\zeta, \beta\rangle, \tag{117}$$

which follows from the fact that

$$\hat{S}^\dagger(\zeta)\hat{D}(\alpha)\hat{S}(\zeta) = \hat{D}(\beta). \tag{118}$$

[8] To show this, use the fact that $\mu^2 - |\nu|^2 = 1$.

[9] which is the definition used in Meystre and Sargent

5.3 How to generate squeezing

5.3.1 $\chi^{(2)}$ nonlinear process

Squeezing is typically generated through the use of some nonlinear medium. A medium with a $\chi^{(2)}$ nonlinear polarisation response can be used to parametrically amplify one quadrature and 'de-amplify' another to create squeezing. In particular, a pump of frequency 2Ω can be used to create a squeezed vacuum of frequency Ω. The effective hamiltonian is

$$\hat{\mathcal{H}}_{\chi^{(2)}} = \hbar\Omega a^\dagger a + 2\hbar\Omega c^\dagger c + \hbar\chi\left(c{a^\dagger}^2 + c^\dagger a^2\right), \tag{119}$$

which describes the process of one c photon (from the pump) begin converted into 2 a photons (and vice versa)[10]. If the pump field has a large intensity, it is often a good approximation to represented it as a constant classical field (the 'undepleted pump approximation'), in which case the unitary operator describing the evolution in the interaction picture has the same form as the squeezing operator, Eq. (102). One insight that this gives us is that squeezing is linked to the creation of photons in pairs.

5.3.2 $\chi^{(3)}$ nonlinear process

Squeezing can also be generated with a $\chi^{(3)}$ medium, either through a pair production process similar to the $\chi^{(2)}$ method, but now with 2 pump photons being annihilated in order to create the squeezed pair, or through what is known as the optical Kerr effect (an intensity dependent refractive index). The effective hamiltonian for the latter is

$$\begin{aligned}\hat{\mathcal{H}}_{\chi^{(3)}} &= \hbar\Omega a^\dagger a + \hbar\chi {a^\dagger}^2 a^2 \\ &= \hbar\left[\Omega + \chi(a^\dagger a - 1)\right] a^\dagger a,\end{aligned} \tag{120}$$

which will cause states with higher photon number to oscillate at a higher frequency. In phase-space this means that the initially symmetric distribution of a coherent state will shear into the elliptic distribution of a squeezed state[11]. The advantage with the Kerr approach is that is does not require any phase-matching between different frequency components.

[10] This process is called *degenerate* parametric amplification, as the pair of photons produced are at the same frequency

[11] For sufficiently high intensity or long interaction times, excessive shearing will turn the ellipse into a crescent and the squeezing effect will vanish. But usually this behaviour is usually beyond experimental reach

6 Quasiprobability distributions

Further reading: M&S13.6

It is fairly common to represent coherent, squeezed and other quantum states pictorially using 'ball and stick' diagrams, with the 'stick' describing the mean of the quadratures and the uncertainty 'ball' (a circle or ellipse) giving us the standard deviations. Such representations can be given rigour by defining distributions using coherent states. As we shall see, although any quantum state can be represented by some sort of distribution in phase space, there are some subtleties over how such a distribution is to be interpreted.

Many of the subtleties arise from the overcomplete or *nonorthogonal* nature of coherent states. For later comparison, first recall that we can expand any state over number states as

$$|\psi\rangle = \sum_n C_n |n\rangle, \tag{121}$$

where $C_n \equiv \langle n|\psi\rangle$, or more generally, the density operator can be expanded as:

$$\hat{\rho} = \sum_{n,n'} C_{n,n'} |n\rangle\langle n'|, \tag{122}$$

where $C_{n,n'} \equiv \langle n|\hat{\rho}|n'\rangle$. Note that the diagonal elements of the representation are positive, with $C_{n,n}$ giving the probability of finding the system in the state $|n\rangle$.

Representing a quantum state as a quasiprobability distribution has other uses than merely visualisation. Such distributions can give important insight into the nonclassical nature of some states of light; as an expansion over coherent states, they can simplify some analytic calculation; and when they can be treated as true probabilities they provide a means of numerically sampling the quantum evolution through stochastic methods.

In our discussion below, we always consider the quantum state of a single optical mode, but the definitions and properties are fairly straightforward to generalise to the multimode case.

6.1 Husimi Q function

The Q function of a state is simply defined as its overlap with a coherent state:

$$\begin{aligned} Q(\alpha) &\equiv \frac{1}{\pi} |\langle\psi|\alpha\rangle|^2 \quad \text{(pure state)} \\ &\equiv \frac{1}{\pi} \langle\alpha|\hat{\rho}|\alpha\rangle \quad \text{(general)}. \end{aligned} \tag{123}$$

The Q function exists for any quantum state and is always positive. In particular:

$$0 \leq Q(\alpha) \leq \frac{1}{\pi}. \tag{124}$$

Like all the phase-space distributions that we consider, the Q function is normalised to one:

$$\begin{aligned}
\int d^2\alpha\, Q(\alpha) &= \frac{1}{\pi}\int d^2\alpha \langle\alpha|\hat{\rho}|\alpha\rangle \\
&= \frac{1}{\pi}\int d^2\alpha \sum_n \langle\alpha|\hat{\rho}|n\rangle\langle n|\alpha\rangle \\
&= \frac{1}{\pi}\int d^2\alpha \sum_n \langle n|\alpha\rangle\langle\alpha|\hat{\rho}|n\rangle \\
&= \sum_n \langle n|\hat{\rho}|n\rangle \\
&= \sum_n p_n = 1 \qquad (125)
\end{aligned}$$

$$(126)$$

It is tempting, but incorrect, to say the $Q(\alpha)$ function gives the probability density of the system being in the state $|\alpha\rangle$. What we can say, however, is that $Q(\alpha)$ gives the correct average value of the amplitude α, since:

$$\begin{aligned}
\int d^2\alpha\, \alpha Q(\alpha) &= \frac{1}{\pi}\int d^2\alpha \langle\alpha|\hat{\rho}\alpha|\alpha\rangle \\
&= \frac{1}{\pi}\int d^2\alpha \sum_n \langle\alpha|\hat{\rho}a|n\rangle\langle n|\alpha\rangle \\
&= \frac{1}{\pi}\int d^2\alpha \sum_n \langle n|\alpha\rangle\langle\alpha|\hat{\rho}a|n\rangle \\
&= \sum_n \langle n|\hat{\rho}a|n\rangle \\
&= \mathrm{Tr}\left[\hat{\rho}a\right] \\
&= \langle a\rangle\,. \qquad (127)
\end{aligned}$$

More generally, we can show that moments of the Q function give the expec-

tation values of *antinormally ordered* functions:

$$\begin{aligned}\int d^2\alpha\, \alpha^{*s}\alpha^r Q(\alpha) &= \frac{1}{\pi}\int d^2\alpha \langle\alpha|\alpha^{*s}\hat{\rho}\alpha^r|\alpha\rangle \\ &= \frac{1}{\pi}\int d^2\alpha \langle\alpha|a^{\dagger s}\hat{\rho}a^r|\alpha\rangle \\ &= \mathrm{Tr}\left[\hat{\rho}a^r a^{\dagger s}\right] \\ &= \left\langle a^r a^{\dagger s}\right\rangle . \end{aligned} \tag{128}$$

The positivity of the Q function makes it useful for plotting visualisations of quantum states. One disadvantage is that not every conceivable Q function corresponds to a valid quantum state. For example, it is easy to imagine a narrow Q function that would not obey the uncertainty principle. It is therefore difficult to draw qualitative conclusions about the quantum state from just the Q function alone.

Some examples[12]:

coherent state $|\alpha_0\rangle$

$$Q(\alpha) = \frac{1}{\pi}e^{-|\alpha-\alpha_0|^2}, \tag{129}$$

i.e. a circularly symmetric gaussian with standard deviation $\sigma_Q = 1/\sqrt{2}$.

number state $|n\rangle$

$$Q(\alpha) = \frac{|\alpha|^{2n}e^{-|\alpha|^2}}{\pi n!}, \tag{130}$$

i.e. a ring centred at the origin with a maximum at radius $|\alpha| = \sqrt{n}$. The mean amplitude is zero, but the mean-square is $\int d^2\alpha\, |\alpha|^2 Q(\alpha) = n + 1$.

thermal state

$$Q(\alpha) = \frac{1}{\pi}\frac{1}{\overline{n}+1}\exp\left[-\frac{|\alpha|^2}{\overline{n}+1}\right], \tag{131}$$

squeezed state $|\alpha_0, \zeta\rangle$, with $\zeta = r$ (real) and $\alpha_0 = X_0 + iY_0$:

$$Q(X_1, X_2) = \frac{1}{\pi \mathrm{cosh} r}\exp\left[-\frac{2(X_1 - X_0)^2}{1+e^{-2r}} - \frac{2(X_2 - Y_0)^2}{1+e^{+2r}}\right] \tag{132}$$

i.e. a two-dimensional gaussian whose width, in this case, in the X_1 direction is narrower than in the X_2 direction.

[12] derive these as exercises

6.2 Glauber-Sudarshan P function

The density operator for a mixture of coherent states has the form:

$$\hat{\rho} = \int d^2\alpha \, P(\alpha)|\alpha\rangle\langle\alpha|, \tag{133}$$

where $P(\alpha)$ is the probability density of being in the state $|\alpha\rangle$. If we are willing to allow P to be negative, then we can represent a wide variety of states in this form, due to the overcompleteness of coherent states. In fact, if we admit singular P functions (e.g. a derivative of a delta-function), then any quantum state can be written in this form. Note that because P can take negative values and could be singular, it cannot be regarded in general as a probability distribution.

Just as for the Q function, the P function gives the correct 'average' of the amplitude α:

$$\int d^2\alpha \, \alpha P(\alpha) = \langle a \rangle, \tag{134}$$

but in contrast to the Q function, the moments of the P function give *normally ordered* correlation functions:

$$\begin{aligned}
\langle a^{\dagger r} a^s \rangle &= \mathrm{Tr}\left[\hat{\rho} a^{\dagger r} a^s\right] \\
&= \mathrm{Tr}\left[a^s \hat{\rho} a^{\dagger r}\right] \\
&= \int d^2\alpha \, P(\alpha) \mathrm{Tr}\left[a^s |\alpha\rangle\langle\alpha| a^{\dagger r}\right] \\
&= \int d^2\alpha \, \alpha^s \alpha^{*r} P(\alpha) \mathrm{Tr}\left[|\alpha\rangle\langle\alpha|\right] \\
&= \int d^2\alpha \, \alpha^s \alpha^{*r} P(\alpha).
\end{aligned} \tag{135}$$

This means that if the P function for a state is everywhere positive, then the state is classical in the sense that all the coherence functions (and hence intensity measurements and photocount statistics) can be described by a fluctuating amplitude. But if P is negative or singular, then this classical explanation fails.

Some examples:

coherent state $|\alpha_0\rangle$

$$P(\alpha) = \delta(\alpha - \alpha_0), \tag{136}$$

i.e. a single point in phase space. The P function is thus a very 'efficient' representation.

number state $|n\rangle$

$$P(\alpha) = \frac{e^{|\alpha|^2}}{n!} \left(\frac{\partial^2}{\partial\alpha\partial\alpha^*}\right)^n \delta(\alpha), \tag{137}$$

which, apart from the vacuum $n = 0$, is a highly singular function.

thermal state

$$P(\alpha) = \frac{1}{\pi \overline{n}} \exp\left[-\frac{|\alpha|^2}{\overline{n}}\right], \tag{138}$$

squeezed state For a squeezed state, the P function is negative in places, which we can demonstrate as follows:

$$\begin{aligned} \int d^2\alpha \, (X_1 - \langle \hat{X}_1 \rangle)^2 P(\alpha) &= \langle : \hat{X}_1 : \rangle^2 - \langle \hat{X}_1 \rangle^2 \\ &= \mathrm{var}(X_1) - 1. \end{aligned} \tag{139}$$

If the X_1 quadrature is squeezed then $\mathrm{var}(X_1) < 1$, i.e. the RHS is less than zero. Since $(X_1 - \langle \hat{X}_1 \rangle)^2 > 0$, the only way for the integral on the LHS to be less than zero is if $P(\alpha)$ is not everywhere positive.

If the P function for a state is known, then the Q function can be obtained via:

$$Q(\alpha) = \frac{1}{\pi} \int d^2\beta \, P(\beta) e^{-|\alpha - \beta|^2}, \tag{140}$$

i.e. by a convolution of the P function with a gaussian.

6.3 Wigner function

Wigner defined his phase-space distribution function for a particle as the quantum equivalent of a joint probability function for position and momentum. He found that one can construct a function $W(x, p)$ that gives the correct marginal distributions, i.e.

$$P_x(x) = \int dp \, W(x, p); \quad P_p(p) = \int dx \, W(x, p), \tag{141}$$

such that

$$\langle \hat{x}^s \rangle = \int dx \, x^s P_x(x); \quad \langle \hat{p}^s \rangle = \int dp \, p^s P_p(p), \tag{142}$$

but that the function $W(x, p)$ itself could be negative, which he regarded as a signature of nonclassical behaviour.

We can similarly define a Wigner function for the quadratures $W(X_1, X_2) = W(\alpha)$ such that:

$$\begin{aligned} \langle \hat{X}_1^s \rangle &= \int dX_1 dX_2 X_1^s W(X_1, X_2) \\ \langle \hat{X}_2^s \rangle &= \int dX_1 dX_2 X_2^s W(X_1, X_2), \end{aligned} \tag{143}$$

which is essentially what we are plotting in phase space when we draw an 'uncertainty ellipse' determined by the standard deviations in X_1 and X_2. In general, the moments of the Wigner function are symmetrically ordered correlation functions:

$$\langle \left\{ a^{\dagger r} a^s \right._{\rm sym} \rangle \;=\; \int d^2\alpha\, \alpha^s \alpha^{*r} W(\alpha). \tag{144}$$

A symmetrically ordered product contains all possible orderings of the operators. For example:

$$\begin{aligned} \left\{ a^\dagger a \right._{\rm sym} &= \frac{1}{2}(a^\dagger a + a a^\dagger) \\ \left\{ a^{\dagger 2} a^2 \right\}_{\rm sym} &= \frac{1}{6}(a^\dagger a^\dagger a a + a^\dagger a a^\dagger a + a^\dagger a a a^\dagger + a a^\dagger a^\dagger a + a a^\dagger a a^\dagger + a a a^\dagger a^\dagger) \end{aligned} \tag{145}$$

Given that symmetric ordering is in some sense half way between normal and antinormal ordering, it is perhaps not surprising the Wigner function is 'half way' between the P and the Q functions. An instance of this is that the Wigner function, like the Q function, can also be written as a convolution of the P function and a gaussian:

$$W(\alpha) = \frac{2}{\pi} \int d^2\beta\, P(\beta) e^{-2|\alpha-\beta|^2}, \tag{146}$$

except that now the gaussian has half the variance. This makes the Wigner function wider than the P function (and thus better behaved) but narrower than the corresponding Q function. Like the Q function, a nonsingular Wigner exists for any quantum state. But unlike the Q function, and like the P function, the Wigner function can be negative. The negativity is usually associated with interference fringes arising from superpositions of coherent states.

Some examples:

coherent state $|\alpha_0\rangle$

$$W(\alpha) = \frac{2}{\pi} e^{-2|\alpha-\alpha_0|^2}, \tag{147}$$

i.e. a circularly symmetric gaussian with standard deviation $\sigma_W = 1/2$.

number state $|n\rangle$ has a Wigner function that oscillates from positive to negative as a function of $|\alpha|$ which confirms its highly nonclassical status:

$$W(\alpha) = \frac{2(-1)^n}{\pi} e^{-2|\alpha|^2} L_n(4|\alpha|^2), \tag{148}$$

where L_n is the n Laguerre polynomial.

thermal state

$$W(\alpha) = \frac{2}{\pi} e^{-2|\alpha|^2/(2\bar{n}+1)} \tag{149}$$

squeezed state $|\alpha_0, \zeta\rangle$, with $\zeta = r$ (real) and $\alpha_0 = X_0 + iY_0$:

$$W(X_1, X_2) = \frac{2}{\pi} \exp\left[-2(X_1 - X_0)^2 e^{-2r} - 2(X_2 - Y_0)^2 e^{+2r}\right], \tag{150}$$

which, like the Q function, is a two-dimensional gaussian and thus everywhere positive.

Squeezed states show up the difference between using the P and the Wigner functions as signatures of nonclassicality. The coherence properties of a squeezed state cannot be replicated by classical light with fluctuating intensity (hence its P functions is not positive definite). But its Wigner function is positive definite and thus constitutes a joint probability distribution for the quadratures, i.e. the quadrature statistics could in principle be replicated by classical random variables. This is an example of how some of the predictions of quantum mechanics can be obtained by *hidden variable* theories. To look for differences between quantum mechanics and local hidden variable theories, one has to generate entangled states that violate a Bell's inequality.

6.4 Characteristic functions

We can give a general expression for the Wigner function of an arbitrary density operator by first defining the *symmetric characteristic function*:

$$\chi_S(\lambda) \equiv \mathrm{Tr}\left[\rho e^{\lambda a^\dagger - \lambda^* a}\right], \tag{151}$$

where λ is a complex number. The Wigner function is then defined as the Fourier transform[13] of the symmetric characteristic function:

$$W(\alpha) = \frac{1}{\pi^2} \int d^2\lambda \, e^{\alpha\lambda^* - \alpha^*\lambda} \chi_S(\lambda). \tag{152}$$

Even though we were able to define the Q and P functions without the use of characteristic functions, it can be useful for some analytic calculations to do

[13] to see that $e^{\alpha\lambda^* - \alpha^*\lambda}$ does indeed give the kernel of the Fourier transform, you may like to express all the complex variables in terms of their real and imaginary parts

so. For that purpose, we define the normally ordered and antinormally ordered characteristic functions to be:

$$\begin{aligned} \chi_N(\lambda) &\equiv \mathrm{Tr}\left[\rho e^{\lambda a^\dagger} e^{-\lambda^* a}\right] \\ \chi_A(\lambda) &\equiv \mathrm{Tr}\left[\rho e^{-\lambda^* a} e^{\lambda a^\dagger}\right], \end{aligned} \tag{153}$$

whose Fourier transforms are, respectively, the P and Q functions:

$$\begin{aligned} P(\alpha) &= \frac{1}{\pi^2}\int d^2\lambda\, e^{\alpha\lambda^* - \alpha^*\lambda}\chi_N(\lambda) \\ Q(\alpha) &= \frac{1}{\pi^2}\int d^2\lambda\, e^{\alpha\lambda^* - \alpha^*\lambda}\chi_A(\lambda). \end{aligned}$$